

7.1 Discrete-Time Control Systems

Discrete-systems: There is one or more impulse series or digital signals in the system.

Types: $\begin{cases} \text{Sampling systems: Discrete Time, Continuous Value} \\ \text{Digital systems: Discrete Time, Quantized Value} \end{cases}$

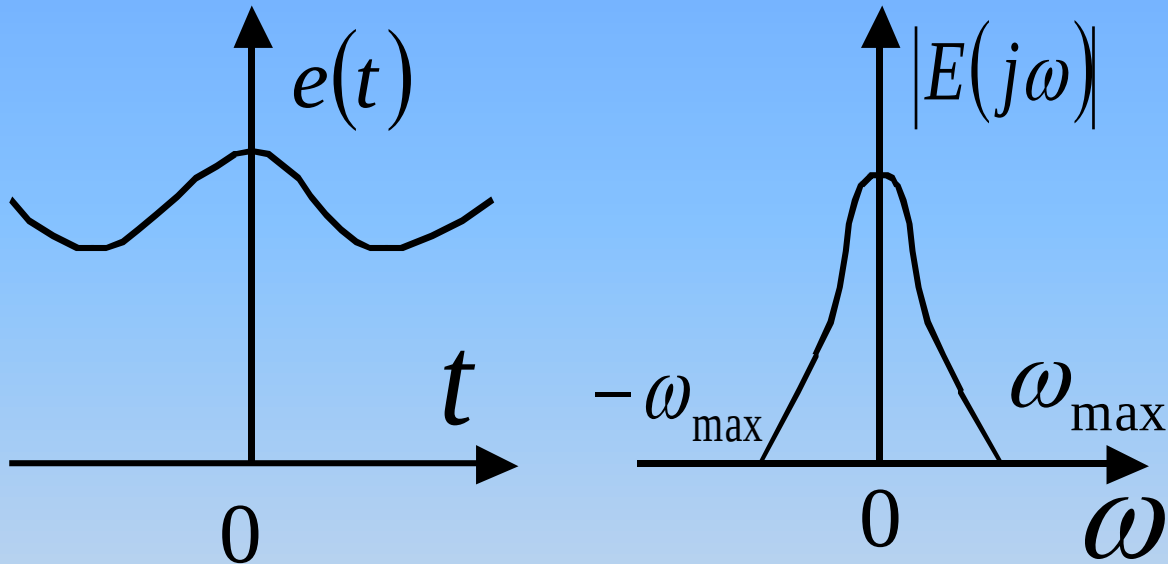
7.2 Signal Sampling and Shannon Theorem

A/D: $\begin{cases} \tau \ll T \\ \text{Enough word length} \end{cases} \quad \begin{cases} \text{It is equivalent to an ideal sampling switch} \\ e^*(t) = e(t) \cdot \delta_T(t) \end{cases}$

D/A: It can be realized by ZOH

Shannon theorem $\omega_s = \frac{2\pi}{T} > 2\omega_h$ or $T < \frac{\pi}{\omega_h}$

The continuous signal and its amplitude spectrum are

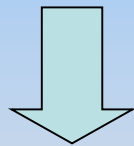


The Fourier-series expansion of $\delta_T(t)$:

$$\delta_T(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_s t}$$

$\omega_s = \frac{2\pi}{T}$ is the sampling freq.

$$c_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \delta_T(t) e^{-jk\omega_s t} dt \stackrel{\text{令 } t=0}{=} \frac{1}{T} \int_{0^-}^{0^+} \delta(t) dt = \frac{1}{T}$$



$$\delta_T(t) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{jk\omega_s t}$$

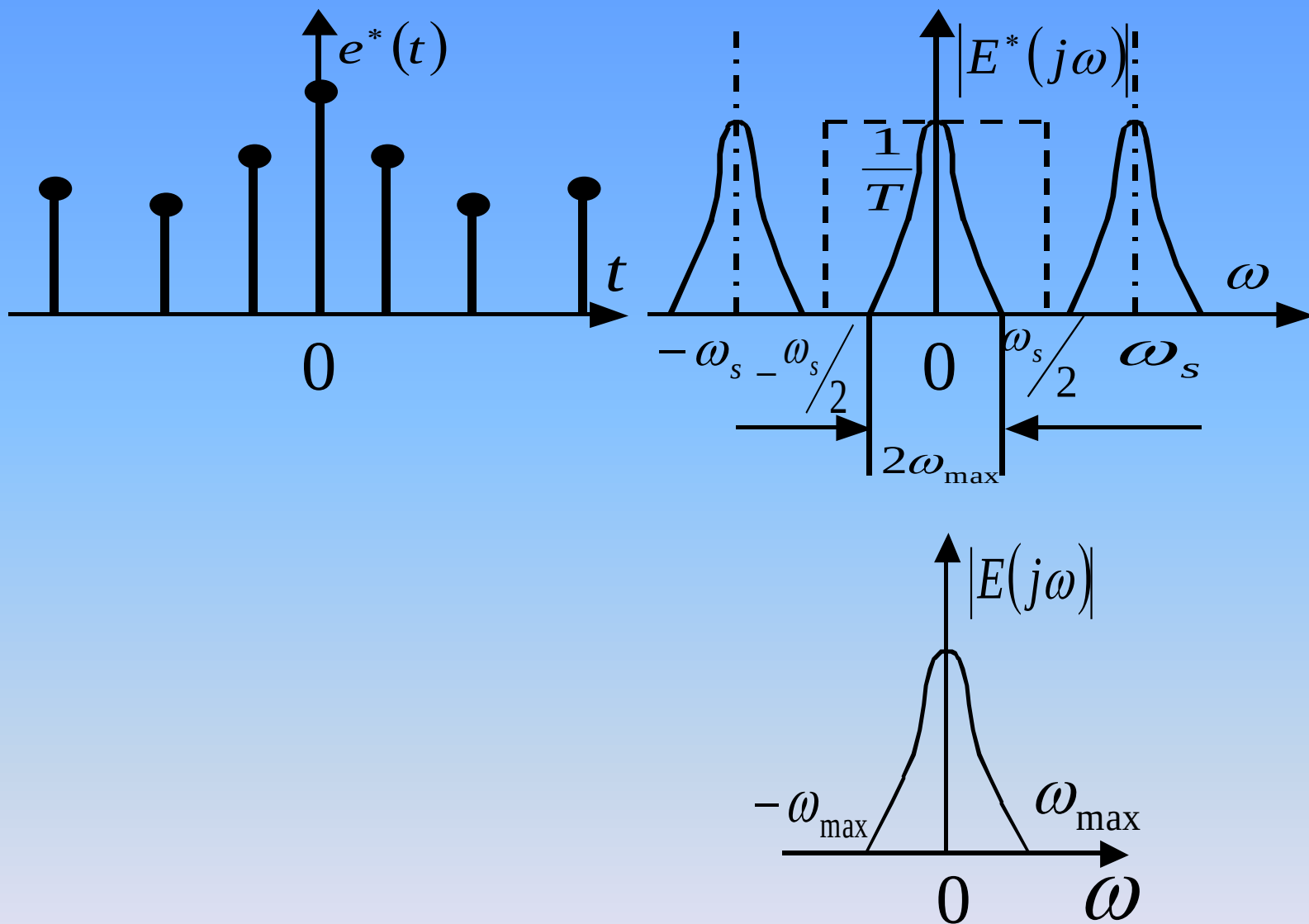
So the sampled signal is

$$e^*(t) = e(t) \cdot \delta_T(t) = \frac{1}{T} \cdot \sum_{k=-\infty}^{\infty} e(t) \cdot e^{jk\omega_s t}$$

which Laplace transform is

$$E^*(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} E[j(\omega + k\omega_s)]$$

where the operator s is replaced by $j\omega$



From the figure above, we can conclude that if $\omega_s > 2\omega_{\max}$

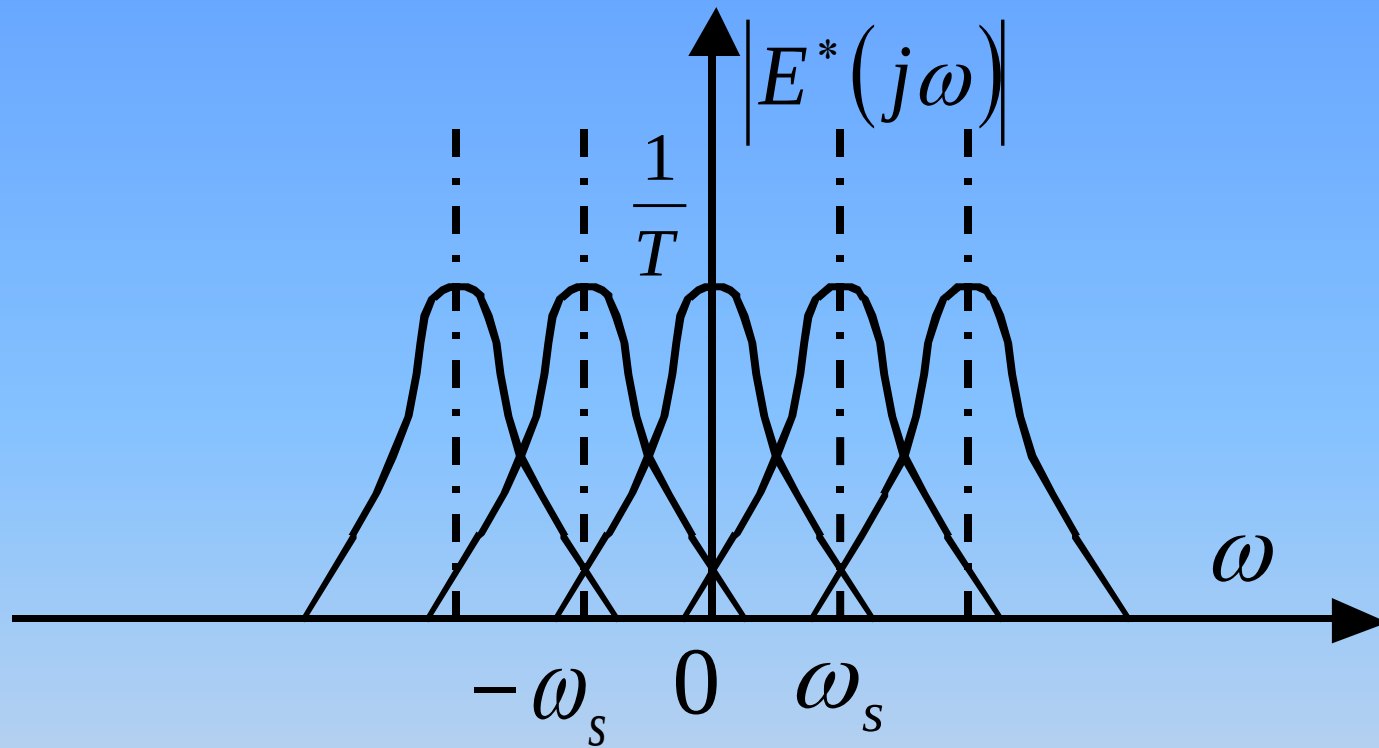
there are no overlap of each component, so the input signal can be recovered approximately. This is called sampling theorem or **Shannon's Theorem**

Shannon's Sampling Theorem:

Let $x(t)$ denote any continuous-time signal having a continuous Fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

Let $x^*(t)$ denote the samples of $x(t)$ at uniform intervals of T seconds. Then $x(t)$ can be exactly reconstructed from its samples $x^*(t)$ if and only if $X(j\omega) = 0$ for all $|\omega| \geq \pi/T$

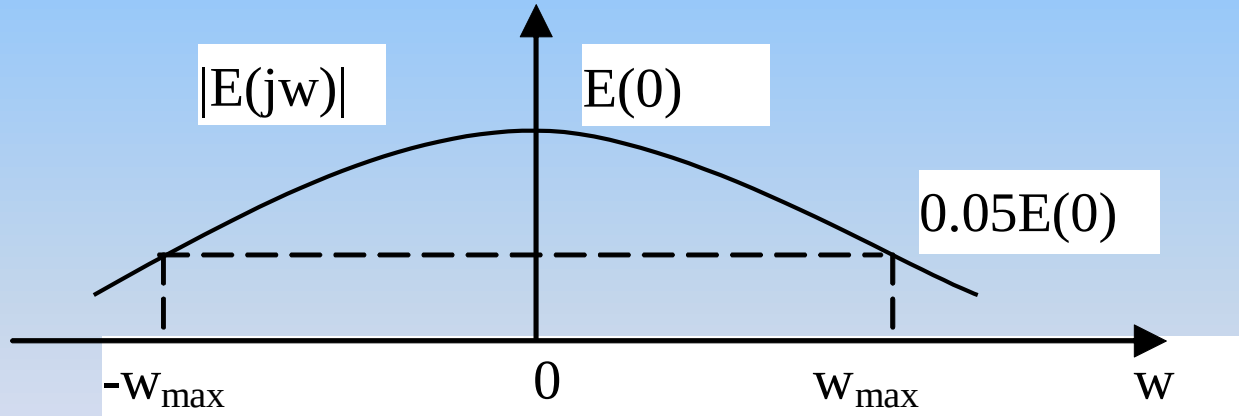


In the figure the input signal can't be recovered.

Problem: the maximum frequency $\omega_{\max}$ is *infinite* for a non-periodic signal!

Then, how could we select the sampling frequency ω_s for it?

Solution:



Example 7-3 Let $e(t)=e^{-t}$, select the sampling frequency according to Shannon's sampling theorem.

Solution.

The Laplace transform of $e(t)$: $E(s) = \frac{1}{s+1}$

The Fourier transform of $e(t)$ is: $E(j\omega) = \frac{1}{j\omega+1}$

The amplitude frequency characteristics:

$$|E(j\omega)| = \frac{1}{\sqrt{\omega^2 + 1}}$$

Assume the maximum frequency of $e(t)$ satisfies: $|E(j\omega_{\max})| = 0.05|E(0)|$

$$\frac{1}{\sqrt{\omega_{\max}^2 + 1}} = 0.05, \quad \omega_{\max} = 20 \text{ rad} / \text{s}$$

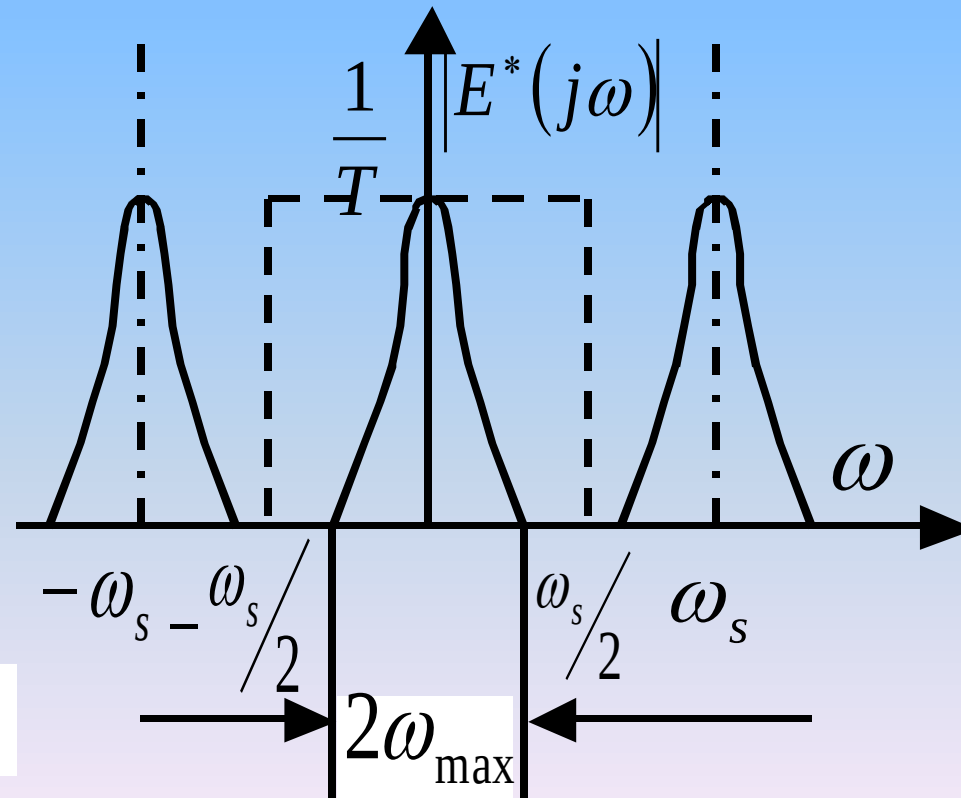
Then according to Shannon's theorem we have

$$\omega_s \geq 2\omega_{\max} = 40 \text{ rad} / \text{s}$$

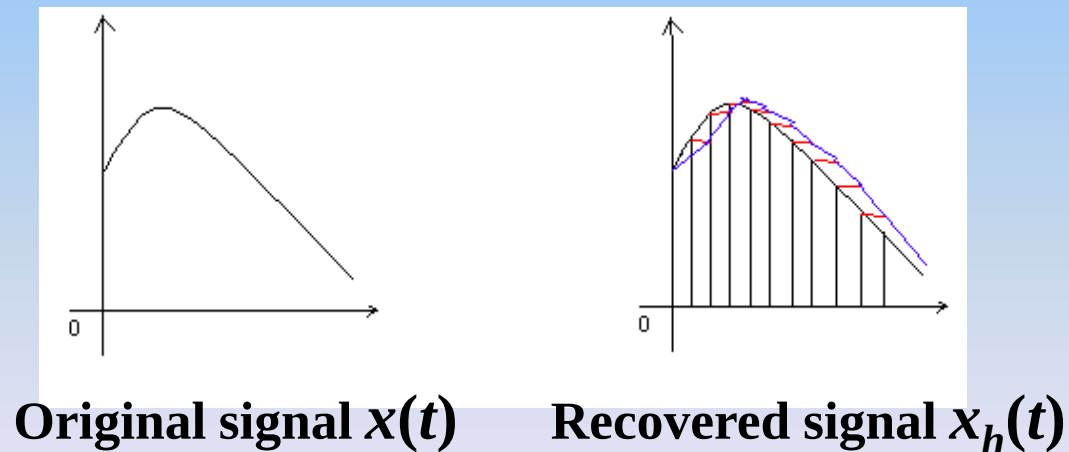
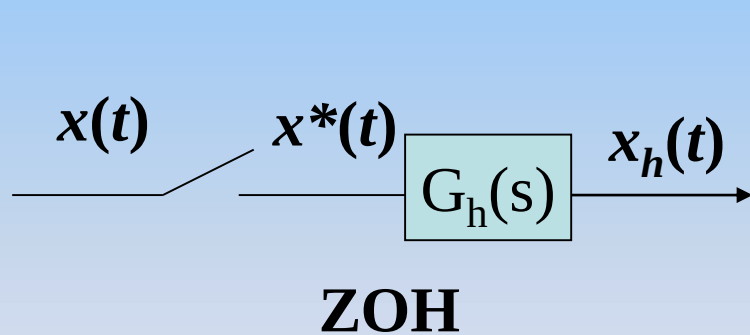
7.3 Signal Recovery and Zero-Order Hold

I. Signal recovery

The ideal filter is illustrated as the dotted line in the figure.



- The most common-used and simplest filter is zero-order hold filter.
- The zero-order hold (ZOH) is a mathematical model of the practical signal reconstruction done by a conventional digital-to-analog converter (DAC). That is, it describes the effect of converting a discrete-time signal to a continuous-time signal by *holding each sample value for one sample interval*.



The original signal $x(t)$ and recovered signal $x_h(t)$ satisfy :

$$x_h(t) = \sum_{k=0}^{\infty} x(kT)(1(t - kT) - 1(t - kT - T))$$

Apply Laplace transform on both sides of the equation, we have

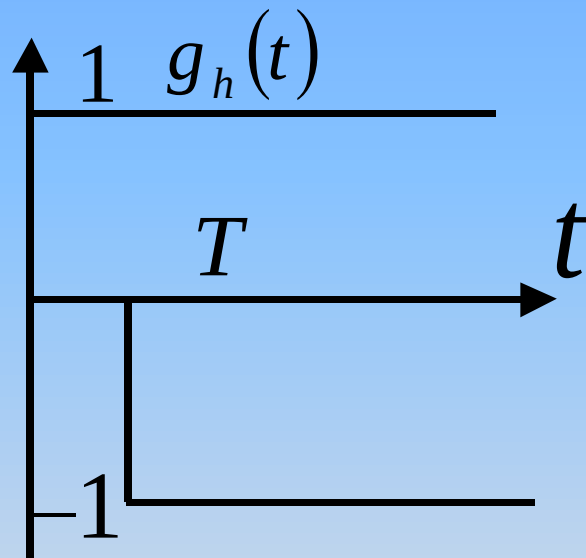
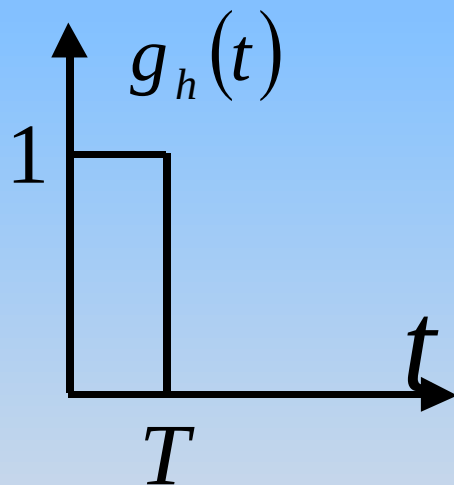
$$x_h(s) = \underbrace{\sum_{k=0}^{\infty} x(kT)e^{-kTs}}_{x^*(s)} \left[\frac{1}{s} - \frac{1}{s} e^{-Ts} \right]$$

Then the ZOH equivalent transition function is

$$\frac{x_h(s)}{x^*(s)} = \frac{1 - e^{-Ts}}{s} = G_h(s)$$

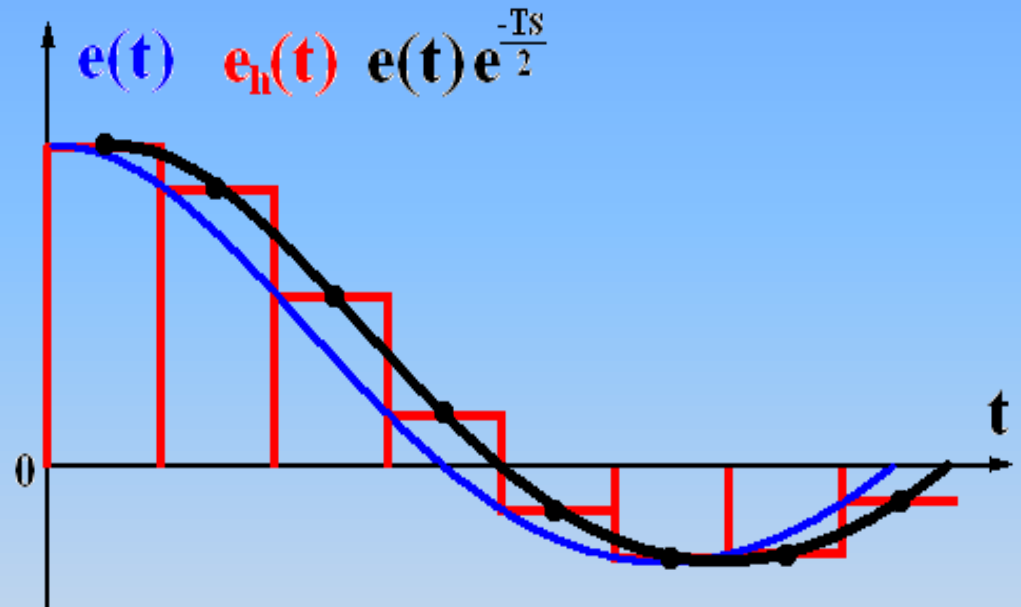
Corresponding time-domain function:

$$g_h(t) = 1(t) - 1(t - T)$$



Effect of zero-order holder on the system

$$G_h(s) = \frac{1 - e^{-Ts}}{s}$$
$$\approx e^{-Ts/2}$$



Analysis of the frequency characteristics of ZOH filter :

$$G_h(j\omega) = \frac{1 - e^{-j\omega T}}{j\omega} = \frac{e^{-\frac{1}{2}j\omega T} (e^{j\frac{1}{2}\omega T} - e^{-j\frac{1}{2}\omega T})}{j\omega} = \frac{2e^{-\frac{1}{2}j\omega T} \left(\frac{e^{j\frac{1}{2}\omega T} - e^{-j\frac{1}{2}\omega T}}{2j} \right)}{\omega}$$

Considering we have, $\sin x = \frac{e^{jx} - e^{-jx}}{2j}$ therefore

$$G_h(j\omega) = \frac{2e^{-\frac{1}{2}j\omega T} \sin(\frac{1}{2}\omega T)}{\omega} \Rightarrow G_h(j\omega) = T \frac{\sin(\frac{\omega T}{2})}{\frac{\omega T}{2}} e^{-\frac{1}{2}j\omega T}$$

$$G_h(j\omega) = \frac{1 - e^{-j\omega T}}{j\omega} = T \cdot \frac{\sin(\omega T/2)}{\omega T/2} \cdot e^{-\frac{j\omega T}{2}}$$

$$\therefore T = \frac{2\pi}{\omega_s}$$

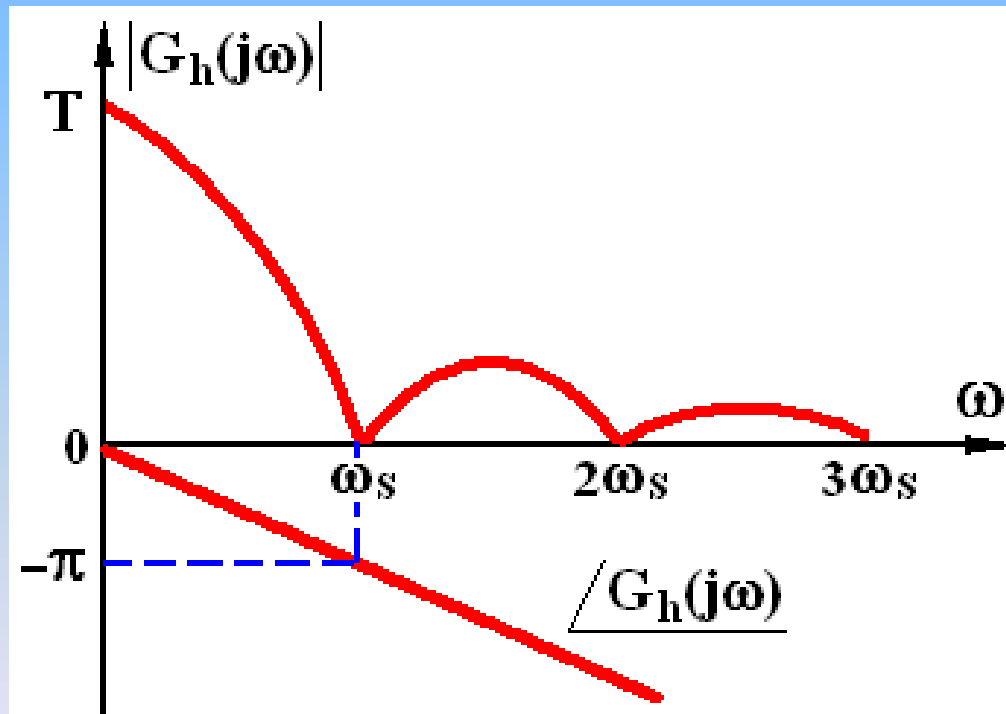
let $S_a(x) = \frac{\sin x}{x}$

then we have

$$G_h(j\omega) = \frac{2\pi}{\omega_s} \cdot S_a(\pi\omega/\omega_s) \cdot e^{-j\frac{\pi\omega}{\omega_s}}$$

Amplitude: $|G_h(j\omega)| = \frac{2\pi}{\omega_s} \cdot |S_a(\pi\omega/\omega_s)|$

Phase angle: $\angle G_h(j\omega) = -\frac{\pi\omega}{\omega_s}$



Note:

- The ZOH filter is not an ideal low-pass filter. *Ripple error* may occur after the filtering.
- There is a *delayed phase angle* when a signal is filtered by a ZOH.

Chapter 7 Analysis and Design of Linear Discrete-Time System (Sampled-data System)

7.1 Introduction

7.2 The Sampling Process and Sampling Theorem

7.3 Signal Recovery and Zero-Order Hold

7.4 Z-Transform and Inverse Z Transform

7.5 Mathematical Models of Discrete-Time Systems

7.6 Dynamic Performance of Discrete-Time Systems

7.7 Digital Control Design for Discrete-Time Systems

7.4 Z-Transform and Inverse Z Transform

7.4.1. Z-transform

Definition:

$$\because E^*(s) = \sum_{k=0}^{+\infty} e(kT) \cdot e^{-kTs}$$

Set

$$z = e^{Ts} \quad s = \frac{1}{T} \ln z$$

$$E(z) = \sum_{k=0}^{\infty} e(kT) \cdot z^{-k}$$

$$E(z) = E^*(s) \big|_{z=e^{Ts}}$$

$$E^*(s) = \sum_{k=0}^{\infty} e(kT) \cdot e^{-kTs}$$

$$z^{-1} = e^{-Ts}$$

$$E(z) = Z[e^*(t)] = E^*(s) \Big|_{z=e^{Ts}} = \sum_{n=0}^{\infty} e(kT) \cdot z^{-k}$$

Rmk: $\left\{ \begin{array}{l} E(z) = Z[e^*(t)] = Z[E(s)] = Z[E^*(s)] = Z[e(t)] \\ \text{The z-transform is only for discrete signal.} \\ E(z) \text{ is only mapping to a unique } e^*(t), \text{ but not a unique } e(t). \end{array} \right.$

7.4.2 Methods of z-Transform

- $\left\{ \begin{array}{l} \text{By the definition.} \\ \text{Partial fraction expansion.} \end{array} \right.$

1、 By the Definition

Example 1 $x_1(t) = 1(t)$ and $x_2(t) = \sum_{k=0}^{\infty} \delta(t - kT)$, obtain $X_1(z)$ and $X_2(z)$.

Solution:

$$X_1(z) = \sum_{k=0}^{\infty} x_1(kT)z^{-k} = 1 + z^{-1} + z^{-2} + \cdots = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$$

$$X_2(s) = \sum_{k=0}^{\infty} e^{-kTs} \quad \xrightarrow{z = e^{Ts}}$$

$$X_2(z) = \sum_{k=0}^{\infty} z^{-k} = 1 + z^{-1} + z^{-2} + \cdots = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$$

Tips: Though $x_1(t)$ and $x_2(t)$ are not same, they may have the same Z-transform.

$$E(z) = \sum_{k=0}^{\infty} e(kT) \cdot z^{-k}$$

Example 2 $e(t) = \sin \omega t = \frac{1}{2j} [e^{j\omega t} - e^{-j\omega t}]$

$$\begin{aligned} E(z) &= \sum_{k=0}^{\infty} \frac{1}{2j} [e^{j\omega kT} - e^{-j\omega kT}] \cdot z^{-k} = \frac{1}{2j} \sum_{k=0}^{\infty} [(e^{j\omega T} z^{-1})^k - (e^{-j\omega T} z^{-1})^k] \\ &= \frac{1}{2j} \left[\frac{1}{1 - e^{j\omega T} z^{-1}} - \frac{1}{1 - e^{-j\omega T} z^{-1}} \right] = \frac{1}{2j} \left[\frac{z}{z - e^{j\omega T}} - \frac{z}{z - e^{-j\omega T}} \right] \\ &= \frac{1}{2j} \cdot \frac{z(e^{j\omega T} - e^{-j\omega T})}{z^2 - (e^{j\omega T} + e^{-j\omega T})z + 1} = \frac{z \sin \omega T}{z^2 - 2 \cos \omega T \cdot z + 1} \end{aligned}$$

Tip: $e^{j\omega t} = \cos \omega t + j \sin \omega t$

Example 3 $e(t) = t$

$$E(z) = \sum_{k=0}^{\infty} e(kT) \cdot z^{-k}$$

Solution. $E(z) = \sum_{k=0}^{\infty} kT \cdot z^{-k} = T \left[z^{-1} + 2z^{-2} + 3z^{-3} + \dots \right]$

$$= Tz \left[z^{-2} + 2z^{-3} + 3z^{-4} + \dots \right]$$

$$= -Tz \left[\frac{d}{dz} z^{-1} + \frac{d}{dz} z^{-2} + \frac{d}{dz} z^{-3} + \dots \right]$$

$$= -Tz \frac{d}{dz} \left[z^{-1} + z^{-2} + z^{-3} + \dots \right]$$

$$= -Tz \frac{d}{dz} z^{-1} \left[1 + z^{-1} + z^{-2} + \dots \right]$$

$$= -Tz \frac{d}{dz} \left[\frac{1}{z} \cdot \frac{1}{1 - z^{-1}} \right] = -Tz \frac{d}{dz} \left[\frac{1}{z - 1} \right] = \frac{Tz}{(z - 1)^2}$$

2. Partial Fraction Expansion

$$E(z) = \sum_{k=0}^{\infty} e(kT) \cdot z^{-k}$$

Example 4 $E(s) = \frac{1}{(s+a)(s+b)}$ Obtain $E(z)$?

Solution:

$$E(s) = \frac{1}{a-b} \cdot \frac{(s+a) - (s+b)}{(s+a)(s+b)} = \frac{1}{a-b} \left[\frac{1}{s+b} - \frac{1}{s+a} \right]$$

$$e(t) = \frac{1}{a-b} [e^{-bt} - e^{-at}]$$

$$E(z) = \frac{1}{a-b} \sum_{k=0}^{\infty} [e^{-bkT} - e^{-akT}] \cdot z^{-k}$$

$$= \frac{1}{a-b} \left[\sum_{k=0}^{\infty} (e^{-bT} \cdot z^{-1})^k - \sum_{k=0}^{\infty} (e^{-aT} \cdot z^{-1})^k \right]$$

$$= \frac{1}{a-b} \left[\frac{1}{1 - e^{-bT} z^{-1}} - \frac{1}{1 - e^{-aT} z^{-1}} \right] = \frac{1}{a-b} \left[\frac{z}{z - e^{-bT}} - \frac{z}{z - e^{-aT}} \right]$$

The z-transform of typical functions

$f(t)$	$F(s)$	$F(z)$
$\delta(t)$	1	1
$f(t)$	$\frac{1}{s}$	$\frac{z}{z-1}$
t	$\frac{1}{s^2}$	$\frac{zT}{(z-1)^2}$
$t^2/2$	$\frac{1}{s^3}$	$\frac{z(z+1)T^2}{2(z-1)^3}$

e^{-at}	$\frac{1}{s+a}$	$\frac{z}{z-e^{-aT}}$
te^{-at}	$\frac{1}{(s+a)^2}$	$\frac{zTe^{-aT}}{(z-e^{-aT})^2}$
$a^{t/T}$	$\frac{1}{s-(1/T)\ln a}$	$\frac{z}{z-a} \quad (a>0)$
$\sin\omega t$	$\frac{\omega}{s^2+\omega^2}$	$\frac{z\sin\omega T}{z^2-2z\cos\omega T+1}$
$\cos\omega t$	$\frac{s}{s^2+\omega^2}$	$\frac{z^2-z\cos\omega T}{z^2-2z\cos\omega T+1}$
$1-e^{-at}$	$\frac{a}{s(s+a)}$	$\frac{z(1-e^{-aT})}{(z-1)(z-e^{-aT})}$
$e^{-at}\sin\omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$	$\frac{ze^{-aT}\sin\omega T}{z^2-2ze^{-aT}\cos\omega T+e^{-2aT}}$
$e^{-at}\cos\omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$	$\frac{z(z-e^{-aT}\cos\omega T)}{z^2-2ze^{-aT}\cos\omega T+e^{-2aT}}$

7.4.3 Properties of z-Transform

1. linear property $Z[a \cdot e_1^*(t) \pm b \cdot e_2^*(t)] = a \cdot E_1(z) \pm b \cdot E_2(z)$

2. Real shifting theorem 实位移定理

① Lag 延时定理 $Z[e(t - nT)] = z^{-n} E(z)$

Proof. LHS = $\sum_{K=0}^{\infty} e(kT - nT) \cdot z^{-k}$

$j = k - n$

$\downarrow$

$$= \sum_{j=-n}^{\infty} e(jT) \cdot z^{-(j+n)} = z^{-n} \sum_{j=0}^{\infty} e(jT) \cdot z^{-j}$$
$$= z^{-n} E(z) = \text{RHS}$$

Left Hand Side (LHS); Right Hand Side (RHS)

2. Real shifting theorem 实位移定理

② Lead 超前定理

$$Z[e(t + nT)] = z^n \left[E(z) - \sum_{k=0}^{n-1} e(kT) \cdot z^{-k} \right]$$

Proof.

$$\begin{aligned} \text{LHS} &= \sum_{k=0}^{\infty} e(kT + nT) \cdot z^{-k} = z^n \sum_{k=0}^{\infty} e(kT + nT) \cdot z^{-(k+n)} \\ &\quad \downarrow j = k + n \\ &= z^n \sum_{j=n}^{\infty} e(jT) \cdot z^{-j} = z^n \left[\sum_{j=0}^{\infty} e(jT) \cdot z^{-j} - \sum_{j=0}^{n-1} e(jT) \cdot z^{-j} \right] \\ &= z^n \left[E(z) - \sum_{k=0}^{n-1} e(kT) \cdot z^{-k} \right] = \text{RHS} \end{aligned}$$

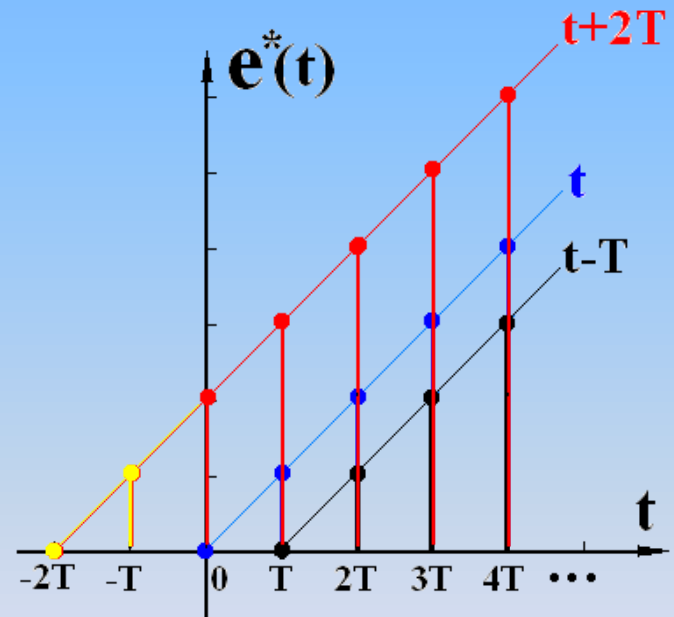
$$Z[e(t + nT)] = z^n \left[E(z) - \sum_{k=0}^{n-1} e(kT) \cdot z^{-k} \right]$$

Example 5 $e(t) = t - T$

$$E(z) = Z[t - T] = z^{-1} Z[t] = z^{-1} \frac{Tz}{(z-1)^2} = \frac{T}{(z-1)^2}$$

Example 6 $e(t) = t + 2T$

$$\begin{aligned} E(z) &= Z[t + 2T] \\ &= z^2 \left\{ Z[t] - \sum_{k=0}^1 kT \cdot z^{-k} \right\} \\ &= z^2 \left[\frac{Tz}{(z-1)^2} - 0 - Tz^{-1} \right] \end{aligned}$$



3. Complex shifting theorem 复位移定理

$$Z[e(t) \cdot e^{\mp at}] = E(z \cdot e^{\pm aT})$$

Proof.

$$\text{LHS} = \sum_{k=0}^{\infty} e(kT) \cdot e^{\mp akT} z^{-k} = \sum_{k=0}^{\infty} e(kT) \cdot (z \cdot e^{\pm aT})^{-k}$$

$$\downarrow \quad z_1 = z \cdot e^{\pm aT}$$

$$= \sum_{k=0}^{\infty} e(kT) \cdot (z \cdot e^{\pm aT})^{-k} = E(z_1) = E(z \cdot e^{\pm akT}) = \text{RHS}$$

Example 7 $e(t) = t \cdot e^{-at}$

$$E(z_1) = Z[t]_{z_1 = z \cdot e^{aT}} = \frac{Tz_1}{(z_1 - 1)^2} = \frac{T(z \cdot e^{aT})}{(z \cdot e^{aT} - 1)^2} = \frac{Tz \cdot e^{-aT}}{(z - e^{-aT})^2}$$

4. Initial-value Theorem

$$\lim_{n \rightarrow 0} e(nT) = \lim_{z \rightarrow \infty} E(z)$$

Proof:

$$\begin{aligned} E(z) &= \sum_{n=0}^{\infty} e(nT) \cdot z^{-n} \\ &= [e(0) + e(1) \cdot z^{-1} + e(2) \cdot z^{-2} + e(3) \cdot z^{-3} + \dots] \end{aligned}$$

$$\lim_{z \rightarrow \infty} E(z) = e(0)$$

Example 8
$$E(z) = \frac{0.792 \cdot z^2}{(z-1)[z^2 - 0.416z + 0.208]}$$

$$e(0) = \lim_{z \rightarrow \infty} E(z) = 0$$

Properties of z-Transform

1. linear property $Z[a \cdot e_1^*(t) \pm b \cdot e_2^*(t)] = a \cdot E_1(z) \pm b \cdot E_2(z)$
2. Real shifting theorem
$$\begin{cases} \text{Lag} & Z[e(t - nT)] = z^{-n} E(z) \\ \text{Lead} & Z[e(t + nT)] = z^n \left[E(z) - \sum_{k=0}^{n-1} e(kT) \cdot z^{-k} \right] \end{cases}$$
3. Complex shifting theorem $Z[e(t) \cdot e^{\mp at}] = E(z \cdot e^{\pm aT})$
4. Initial-value theorem $\lim_{n \rightarrow 0} e(nT) = \lim_{z \rightarrow \infty} E(z)$
5. Final-value theorem $\lim_{n \rightarrow \infty} e(nT) = \lim_{z \rightarrow 1} (z - 1) \cdot E(z)$
6. Convolution theorem $c^*(t) = e^*(t) * g^*(t) \Rightarrow C(z) = E(z) \cdot G(z)$

7.4.4 Inverse z-Transform

$$Z^{-1}[X(z)] = x(nT)$$

Tips:

Inverse Z-transform can only provide discrete-time signal $x^*(t)$, instead of continuous signal $x(t)$ 。

{	Long Division (长除法)	Expansion of $\frac{E(z)}{z}$
	Partial-Fraction expansion	
	Residue (留数法)	

$$e(nT) = \sum \text{Res} \left[E(z) \cdot z^{n-1} \right]$$

1. Long Division (长除法)

$$E(z) = \frac{b_0 z^m + b_1 z^{m-1} + \cdots + b_m}{a_0 z^n + a_1 z^{n-1} + \cdots + a_n}$$

Numerator is divided by denominator ,we get

$$E(z) = c_0 + c_1 z^{-1} + \cdots + c_k z^{-k} + \cdots = \sum_{k=0}^{\infty} c_k z^{-k} = \sum_{k=0}^{\infty} e(kT) z^{-k}$$

$$e^*(t) = c_0 \delta(t) + c_1 \delta(t-T) + \cdots + c_k \delta(t-kT) + \cdots$$

Example: $E(z) = \frac{10z}{(z-1)(z-2)}$, obtain $e^*(t)$.

Solution: $E(z) = \frac{10z}{z^2 - 3z + 2} = 10z^{-1} + 30z^{-2} + 70z^{-3} + 150z^{-4} + \dots$

$$\begin{array}{r}
 10z^{-1} + 30z^{-2} + 70z^{-3} + 150z^{-4} \dots \\
 \hline
 z^2 - 3z + 2 \overline{) 10z} \\
 \underline{10z - 30z^0 + 20z^{-1}} \\
 30z^0 - 20z^{-1} \\
 \underline{30z^0 - 90z^{-1} + 60z^{-2}} \\
 70z^{-1} - 60z^{-2} \\
 \underline{70z^{-1} - 210z^{-2} + 140z^{-3}} \\
 150z^{-2} - 140z^{-3} \\
 \dots
 \end{array}$$

$$\begin{aligned}
 e^*(t) = & 10\delta(t - T) \\
 & + 30\delta(t - 2T) \\
 & + 70\delta(t - 3T) \\
 & + 150\delta(t - 4T) \\
 & + \dots
 \end{aligned}$$

Example

$$F(z) = \frac{z}{(z-2)(z-3)}, \text{ obtain } f^*(t).$$

Solution:

$$\text{Because } F(z) = \frac{z}{z^2 - 5z + 6} = \frac{z^{-1}}{1 - 5z^{-1} + 6z^{-2}}$$

By long-division, we get that

$$F(z) = z^{-1} + 5z^{-2} + 19z^{-3} + 65z^{-4} + \dots$$

Thus

$$f(0) = 0, \quad f(T) = 1, \quad f(2T) = 5, \quad f(3T) = 19, \quad f(4T) = 65, \dots$$


Then

$$f^*(t) = \delta(t-T) + 5\delta(t-2T) + 19\delta(t-3T) + 65\delta(t-4T) + \dots$$

2. Partial fraction expansion

Note: here, we expand $\frac{X(z)}{z}$, instead of $X(z)$.

$$\frac{X(z)}{z} = \sum_{i=1}^n \frac{A_i}{z - z_i}$$

Consider

$$X(z) = \frac{b_0 z^m + b_1 z^{m-1} + \cdots + b_{m-1} z + b_m}{a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n}$$

Then

$$X(z) = \frac{b_0 z^m + b_1 z^{m-1} + \cdots + b_{m-1} z + b_m}{a_0 \prod_{i=1}^n (z - z_i)}$$

If there is no repeated root for the denominator, it generates

$$X(z) = z \left(\frac{A_1}{z - z_1} + \frac{A_2}{z - z_2} + \cdots + \frac{A_n}{z - z_n} \right)$$

其中系数 A_i 可由式决定:

$$A_i = \left[(z - z_i) \frac{X(z)}{z} \right] \Big|_{z=z_i}$$

Example Consider

$$F(z) = \frac{z}{(z-1)(z-e^{-T})}$$

Obtain $f^*(t)$.

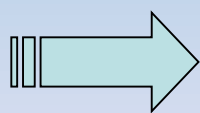
Solution:

$$\frac{F(z)}{z} = \frac{K_1}{z-1} + \frac{K_2}{z-e^{-T}}$$

$$K_1 = \lim_{z \rightarrow 1} \left(\frac{z-1}{z} \right) F(z) = \frac{1}{1-e^{-T}}$$

$$F(z) = \frac{1}{1-e^{-T}} \left(\frac{z}{z-1} - \frac{z}{z-e^{-T}} \right)$$

$$K_2 = \lim_{z \rightarrow e^{-T}} \left(\frac{z-e^{-T}}{z} \right) F(z) = -\frac{1}{1-e^{-T}}$$



$$f(nT) = \frac{1}{1-e^{-T}} (1 - e^{-nT})$$

$$f^*(t) = \frac{1}{1-e^{-T}} \sum_{k=0}^{+\infty} (1 - e^{-kT}) \delta(t - kT)$$

3、Residue(留数法)

$$F(z) = \sum_{k=0}^{+\infty} f(kT)z^{-k}$$

$$F(z)z^{m-1} = \sum_{k=0}^{+\infty} f(kT)z^{m-k-1}$$

Γ Encircle all the poles of $F(z)z^{k-1}$

$$\oint_{\Gamma} F(z)z^{m-1}dz = \oint_{\Gamma} \left[\sum_{k=0}^{+\infty} f(kT)z^{m-k-1} \right] dz$$

$$\oint_{\Gamma} F(z)z^{m-1}dz = \sum_{k=0}^{+\infty} f(kT) \oint_{\Gamma} z^{m-k-1} dz$$

When $m=k$,

$$f(kT) = \sum_{i=1}^n \text{res}[F(z)z^{k-1}, z_i]$$

$z_i, i = 1, 2, \dots, n$ are all the poles of $F(z)z^{k-1}$

$$\text{Res}\left[z^{(k-1)}x(z)\right]=\lim_{z\rightarrow z_i}\frac{1}{(r-1)!}\frac{d^{r-1}}{dz^{r-1}}\left[(z-z_i)^r z^{k-1}x(z)\right]$$

其中 $\text{Res}[\]$ 表示函数的留数, r 为极点的阶数。

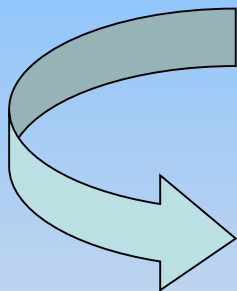
Example For

$$F(z)=\frac{10z}{(z-1)(z-2)}$$

Obtain its inverse z-transform by residue Method.

Solution: $F(z)z^{k-1} = \frac{10z^k}{(z-1)(z-2)}$

Poles $z_1 = 1$ and $z_2 = 2$, and


$$\begin{aligned} \text{res}[F(z)z^{k-1}, 1] &= \lim_{z \rightarrow 1} (z-1)F(z)z^{k-1} = -10 \\ \text{res}[F(z)z^{k-1}, 2] &= \lim_{z \rightarrow 2} (z-2)F(z)z^{k-1} = 10 \cdot 2^k \end{aligned}$$

Then $f(kT) = 10(2^k - 1) \quad (k = 0, 1, 2, \dots)$

Example 11 $E(z) = \frac{z^2}{(z-0.8)(z-0.1)}$ Obtain $e^*(t)$. (PFE & Residue)

PFE: $\frac{E(z)}{z} = \frac{z}{(z-0.8)(z-0.1)} = \frac{C_1}{(z-0.8)} + \frac{C_2}{(z-0.1)}$

$$C_1 = \lim_{z \rightarrow 0.8} \frac{z}{(z-0.1)} = \frac{8}{7} \quad C_2 = \lim_{z \rightarrow 0.1} \frac{z}{(z-0.8)} = \frac{-1}{7}$$

$$= \frac{8/7}{(z-0.8)} - \frac{1/7}{(z-0.1)}$$

$$E(z) = \frac{8}{7} \cdot \frac{z}{(z-0.8)} - \frac{1}{7} \cdot \frac{z}{(z-0.1)}$$

$$e(t) = (8 \times 0.8^{\frac{t}{T}} - 0.1^{\frac{t}{T}}) / 7 \quad e(nT) = (8 \times 0.8^n - 0.1^n) / 7$$

$$e^*(t) = \sum_{n=0}^{\infty} [(8 \times 0.8^n - 0.1^n) / 7] \cdot \delta(t - nT)$$

Example 12 $E(z) = \frac{5}{(z-a)^2}$ Obtain $e^*(t)$. (Residue)

Solution.

$$e(nT) = \sum \text{Res} \left[E(z) \cdot z^{n-1} \right] = \text{Res}_{z=a} \left[\frac{5}{(z-a)^2} \cdot z^{n-1} \right]$$

$$e(nT) = \frac{1}{(2-1)!} \lim_{z \rightarrow a} \frac{d}{dz} \left[(z-a)^2 \frac{5 \cdot z^{n-1}}{(z-a)^2} \right]$$

$$= \lim_{z \rightarrow a} \frac{d}{dz} [5 \cdot z^{n-1}]$$

$$= 5 \cdot \lim_{z \rightarrow a} [(n-1) \cdot z^{n-2}]$$

$$= 5 \cdot (n-1) \cdot a^{n-2}$$

$$e^*(t) = \sum_{n=0}^{\infty} (5(n-1) \cdot a^{n-2}) \cdot \delta(t - nT)$$

7.4.5 Limitations of z-Transform

- (1) only shows the information of samples;
- (2) In some cases, the continuous signal may jump on the sampling point.

